How-To Library

Informatica

Creating Column Profiles on LDAP Data Objects

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Abstract

You can use Informatica Developer to read data from Lightweight Directory Access Protocol (LDAP) directory servers. You can create column profiles on the LDAP data sources and view the profile results to find the content, quality, and structure of the data sources. This article is written for database administrators and developers and assumes that you have basic knowledge of LDAP and Informatica Data Quality product.

Supported Versions

- Data Quality 10.0

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Overview

You can use LDAP to access X.500-based directory services. LDAP defines a method to access and update information in a directory. You can run column profiles on the LDAP data objects to view the data quality issues in the LDAP data sources.

You can view the quality of data such as structure, content, null values, duplicate values, and patterns in the LDAP data sources. To identify the data quality issues in LDAP data sources, create and run column profiles, and view the profile results to analyse the data. Before you create the column profiles, create a connection in Informatica Administrator or the Developer tool to import LDAP metadata. After you create a connection, create data objects in the Developer tool. The LDAP data objects represent the metadata for LDAP entries. Create column profiles on the data objects in Informatica Analyst or the Developer tool.

For more information about Informatica integration with LDAP directory server, see the *PowerExchange for LDAP User Guide*.

Business Example

You are a database administrator and you manage the databases of your company. Your company had a recent acquisition and you want to synchronize the data from the third-party LDAP directory service to the Microsoft Active Directory of your organization. Before you synchronize the list of employees, roles provisioned to users, profile information, contacts, and calendar resources to Active Directory, you want to view the data quality of the Employee table.

To view the content, structure, and quality of data in the Employee table, perform the following tasks:

1. Create a connection in Informatica Developer or Informatica Administrator.

2. Create an LDAP data object in the Developer tool.
3. Create a read operation for the data object in the Developer tool.
4. Create and run the column profile on the read operation in the Developer tool or the Analyst tool.
5. View the profile results in the Developer tool or Analyst tool.

After analysing the profile results, you can take appropriate steps and synchronize the data.

LDAP Connection

Create a LDAP connection to import LDAP metadata. Configure an LDAP connection before the Data Integration Service can read data from the LDAP sources or write data to the LDAP targets. You can create an LDAP connection in the Developer tool, the Administrator tool, or through the infacmd isp command.

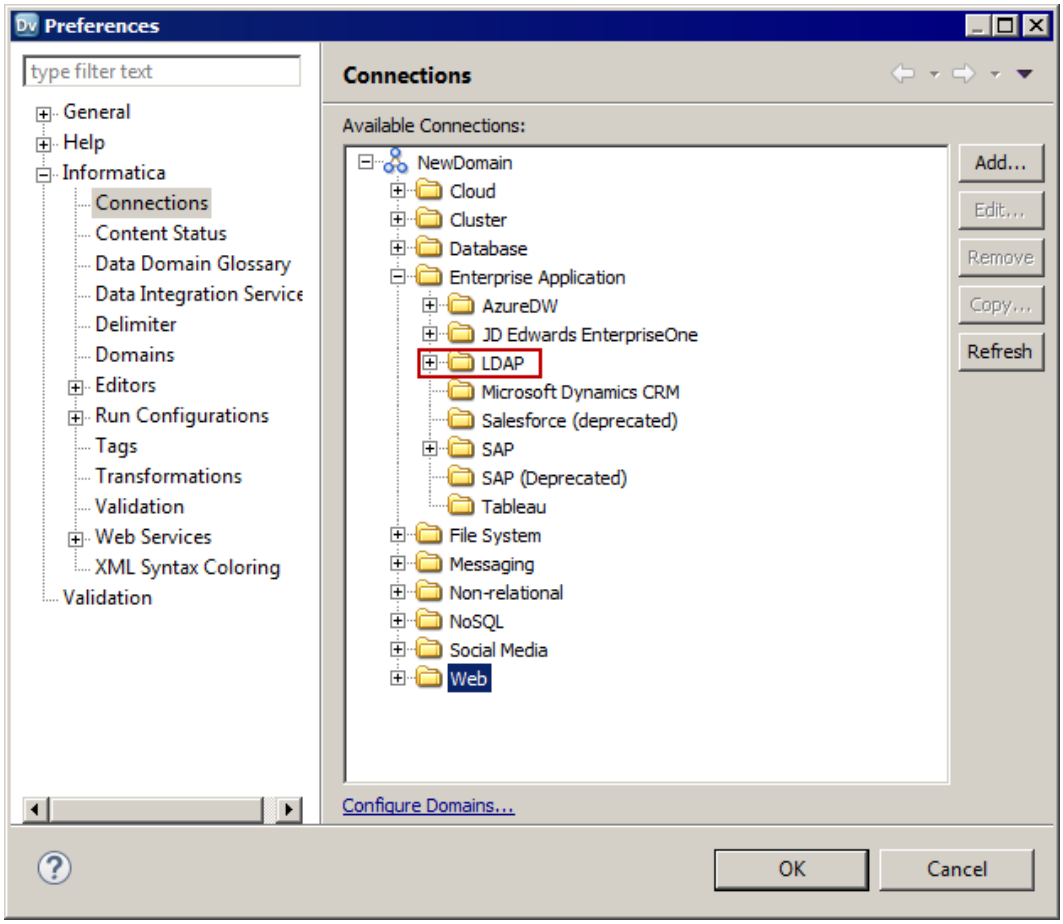
Creating an LDAP Connection in the Developer Tool

Create an LDAP connection in the Developer tool to import LDAP metadata. After you create an LDAP connection, you can create LDAP data objects.

1. In the Developer tool, click **Windows** > **Preferences** option.

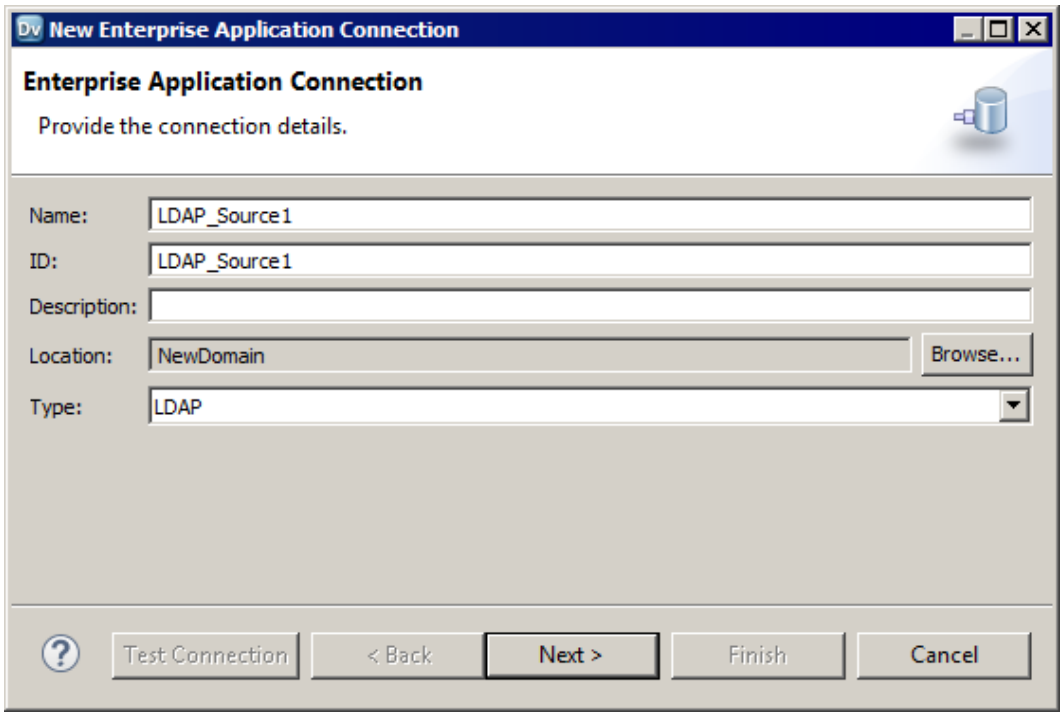
   The **Preferences** dialog box appears.
2. Click **Informatica** > **Connections**.
3. In the **Connections** section, expand the domain.

   The following sample image shows the **Preferences** dialog box and all the available connections in the domain:



4. Select **Enterprise Applications** > **LDAP**, and click **Add**.

   The following sample image shows the **New Enterprise Application Connection** dialog box:



5. In the **Enterprise Application Connection** dialog box, enter the following details:

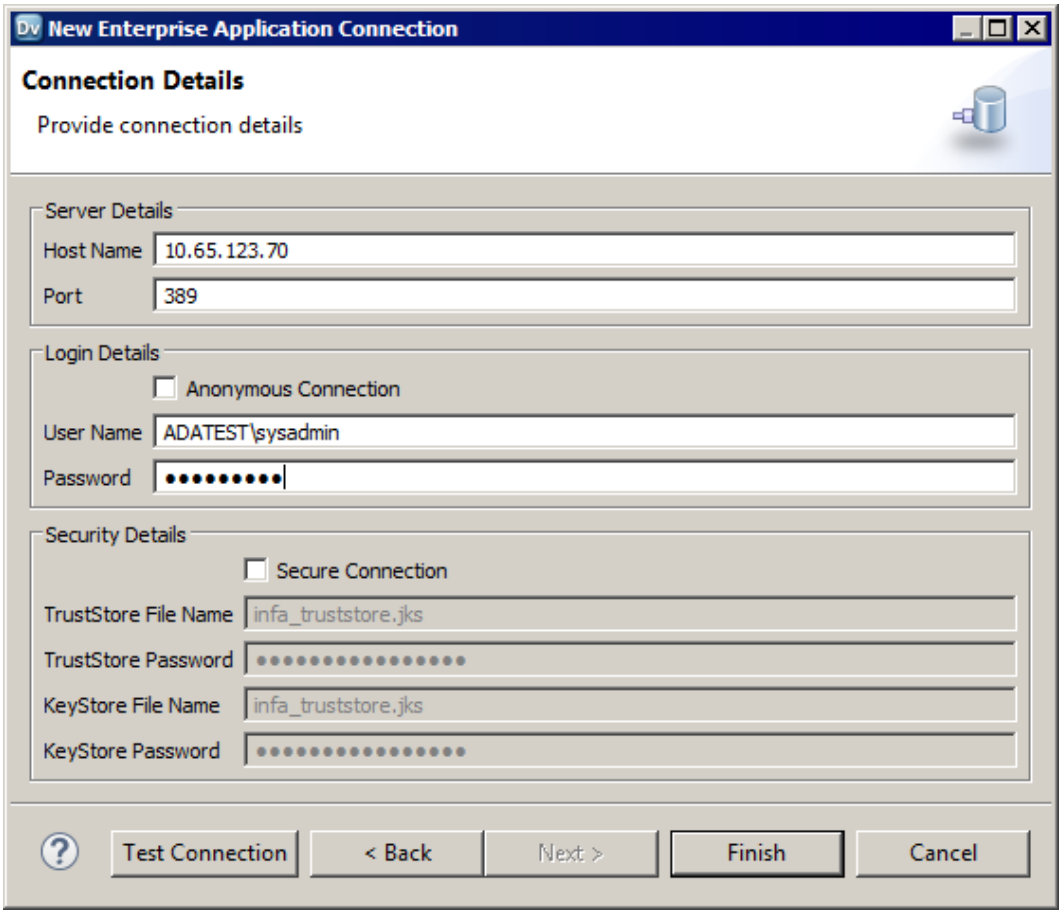
Option	Description
Name	The name of the connection. The name is not case sensitive and must be unique within the domain. It cannot exceed 128 characters, contain spaces, or contain the following special characters: ~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /
ID	The string that the Data Integration Service uses to identify the connection. The ID is not case sensitive. It must be 255 characters or less and must be unique in the domain. You cannot change this property after you create the connection. Default value is the connection name.
Description	Optionally, enter the description of the connection. The description cannot exceed 765 characters.
Location	The Informatica domain where you want to create the connection.
Type	The connection type. Select LDAP.

6. Click **Next**.
7. In the **Connection Details** dialog box, enter the following details:

Option	Description
Hostname	LDAP directory server host name. Default is localhost.
Port	LDAP directory server port number. Default is 389.

Option	Description
Anonymous Connection	Establishes an anonymous connection with the LDAP directory server. Select anonymous connection to access a directory server as an anonymous user without authentication. **Note:** You cannot establish an anonymous connection with Active Directory.
Username	The LDAP user name to connect to the LDAP directory server.
Password	The password to connect to the LDAP directory server.
Secure Connection	Establishes a secure connection with the LDAP directory server through the TLS protocol.
Truststore File Name	The file name of the truststore that contains the TLS certificate to establish a secure connection with the LDAP directory server. Default is `infa_truststore.jks`. Required if you select Secure Connection. Contact the LDAP Administrator for the truststore file name and password.
Truststore Password	The password for the truststore file that contains the SSL certificate.
KeyStore File Name	The file name of the keystore that contains the keys and certificates required to establish a secure communication with the LDAP directory server. Required if you select Secure Connection. Contact the LDAP Administrator for the keystore file name and password.
KeyStore Password	The password for the keystore file required for secure communication.

The following sample image shows the **Connection Details** dialog box:



8. Click **Test Connection**.

   The **Connection successful** message appears.

9. Click **Finish**.

   The LDAP connection appears in the list of connections under the **Connections** section in the **Preferences** dialog box.

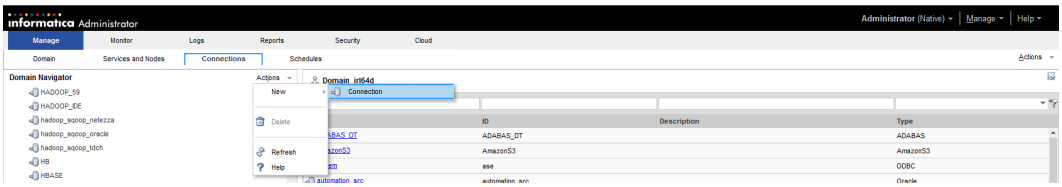
10. Click **OK** to close the **Preferences** dialog box.

Creating an LDAP Connection in the Administrator Tool

Create an LDAP connection in the Administrator tool before you create LDAP data objects.

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Connections** view.
3. In the **Domain Navigator**, select the domain.
4. In the **Domain Navigator**, click **Actions** > **New** > **New Connection**.

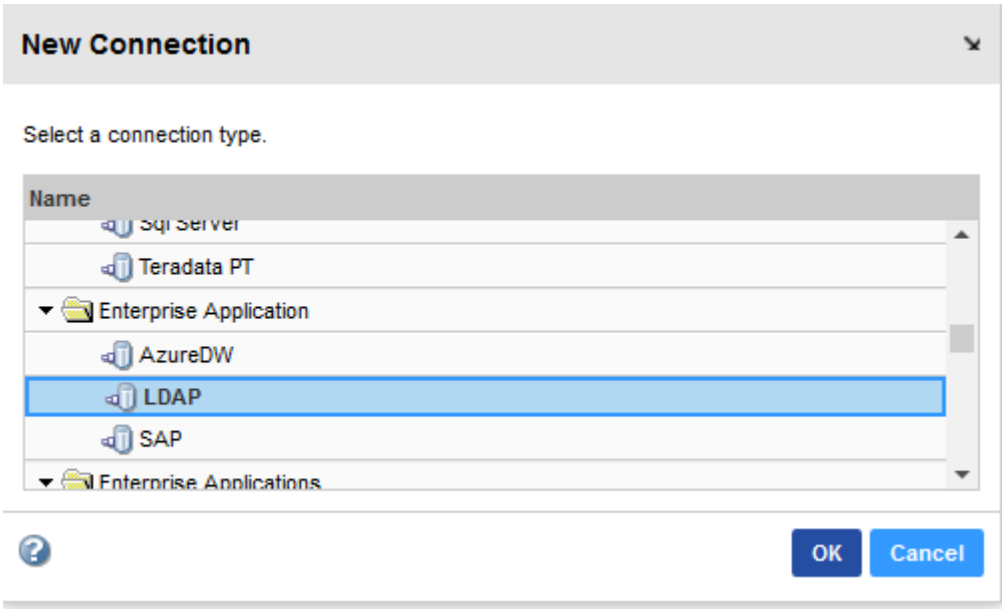
   The following sample image shows the **Domain Navigator** view in the Administrator tool:



The **New Connection** dialog box appears.

5. In the **New Connection** dialog box, select **LDAP**, and click **OK**.

   The following sample image shows the **New Connection** dialog box:



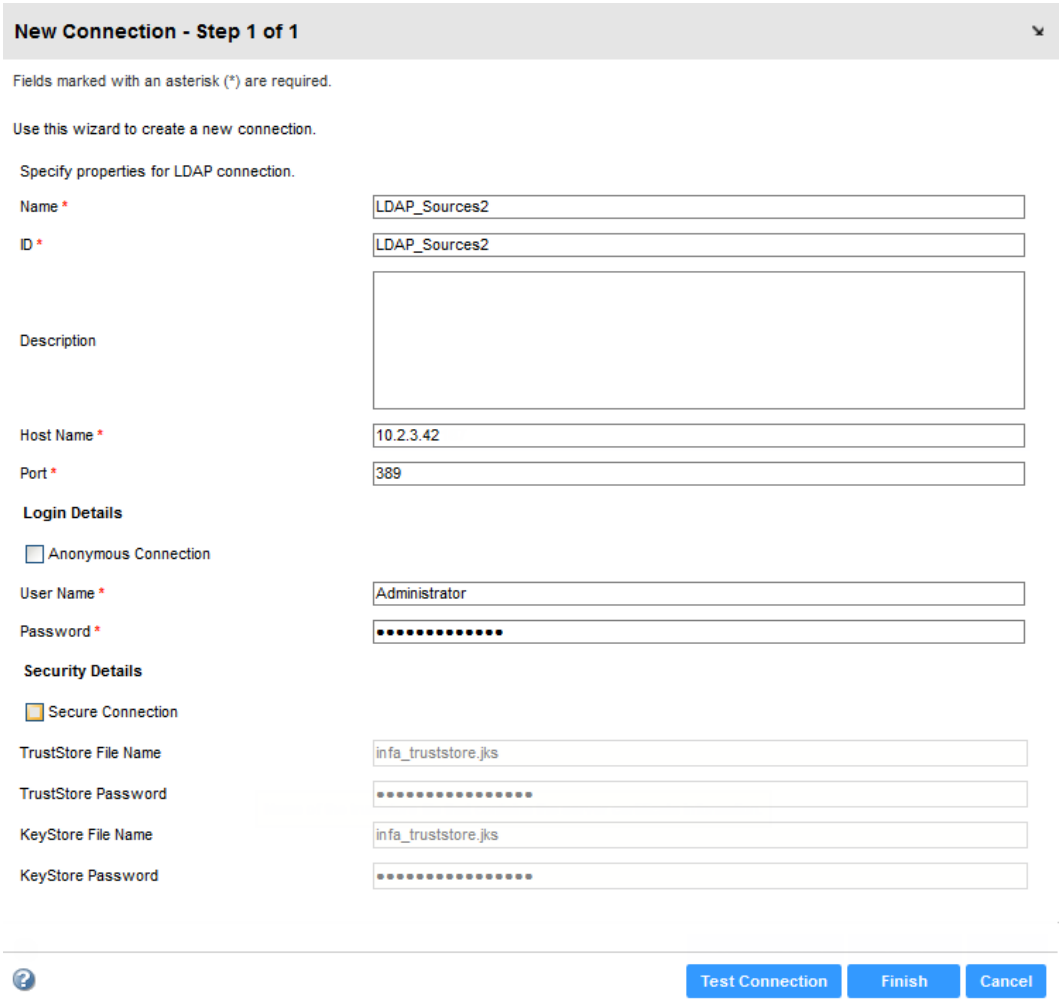
   The **New Connection - Step 1 of 1** dialog box appears.

6. In the **New Connection - Step 1 of 1** dialog box, enter the following details:

Option	Description
Name	The name of the connection. The name is not case sensitive and must be unique within the domain. It cannot exceed 128 characters, contain spaces, or contain the following special characters: ~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /
ID	The string that the Data Integration Service uses to identify the connection. The ID is not case sensitive. It must be 255 characters or less and must be unique in the domain. You cannot change this property after you create the connection. Default value is the connection name.
Description	Optionally, enter the description of the connection. The description cannot exceed 765 characters.
Hostname	LDAP directory server host name. Default is localhost.
Port	LDAP directory server port number. Default is 389.

Option	Description
Anonymous Connection	Establishes an anonymous connection with the LDAP directory server. Select anonymous connection to access a directory server as an anonymous user without authentication. **Note:** You cannot establish an anonymous connection with Active Directory.
Username	The LDAP user name to connect to the LDAP directory server.
Password	The password to connect to the LDAP directory server.
Secure Connection	Establishes a secure connection with the LDAP directory server through the TLS protocol.
Truststore File Name	The file name of the truststore that contains the TLS certificate to establish a secure connection with the LDAP directory server. Default is `infa_truststore.jks`. Required if you select Secure Connection. Contact the LDAP Administrator for the truststore file name and password.
Truststore Password	The password for the truststore file that contains the SSL certificate.
KeyStore File Name	The file name of the keystore that contains the keys and certificates required to establish a secure communication with the LDAP directory server. Required if you select Secure Connection. Contact the LDAP Administrator for the keystore file name and password.
KeyStore Password	The password for the keystore file required for secure communication.

The following sample image shows the **New Connection - Step 1 of 1** dialog box:



7. Click **Test Connection**.

   The **Connection successful** message appears.

8. Click **Finish**.

   The LDAP connection appears in the list of connections for the domain in the **Domain Navigator**.

LDAP Data Objects

The LDAP data objects represent metadata for LDAP entries. An LDAP source or target can contain attributes that have single or multiple values. The attributes can be optional or mandatory. A single-valued attribute can store one value at a time, and a multivalued attribute can store more than one value at a time.

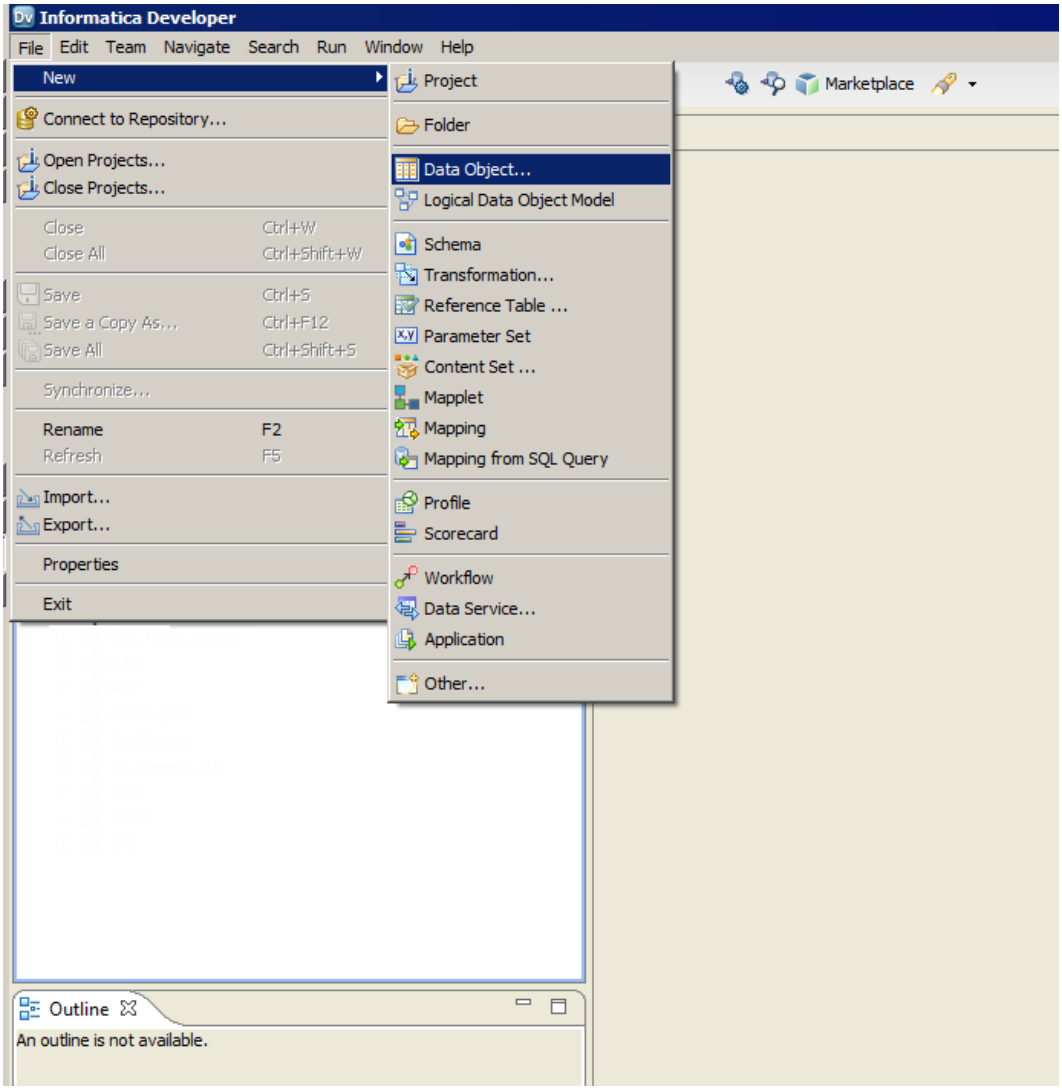
After you configure an LDAP connection, create an LDAP data object in the Developer tool. Create a read operation for the LDAP data object to read data from the LDAP directory server.

Importing an LDAP Data Object in the Developer Tool

Import an LDAP data object in the Developer tool. After you create a data object, you can create a column profile on the data object.

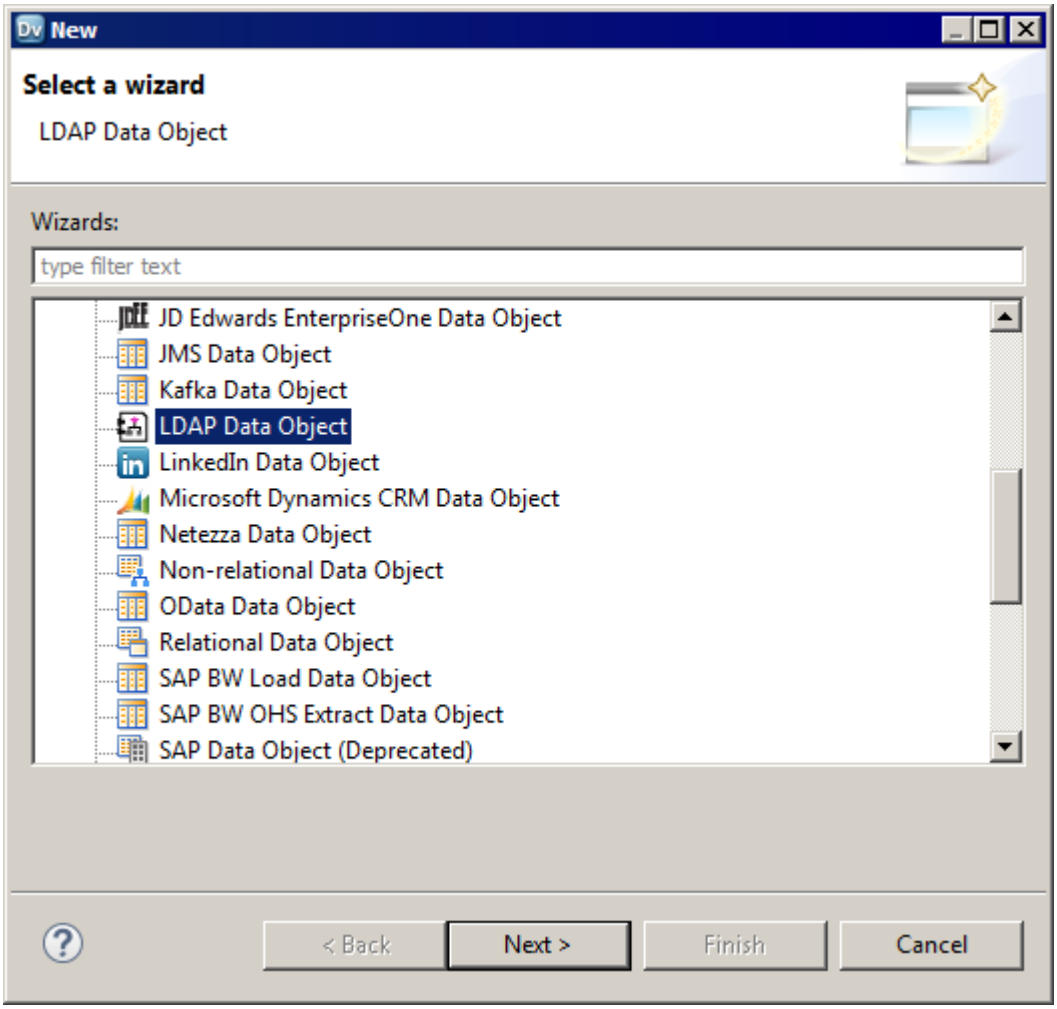
1. In the Developer tool, select a project or folder in the **Object Explorer** view.
2. Click **File** > **New** > **Data Object**.

   The following image shows the option to create a data object:



3. In the **Select a wizard** dialog box, select **LDAP Data Object**, and click **Next**.

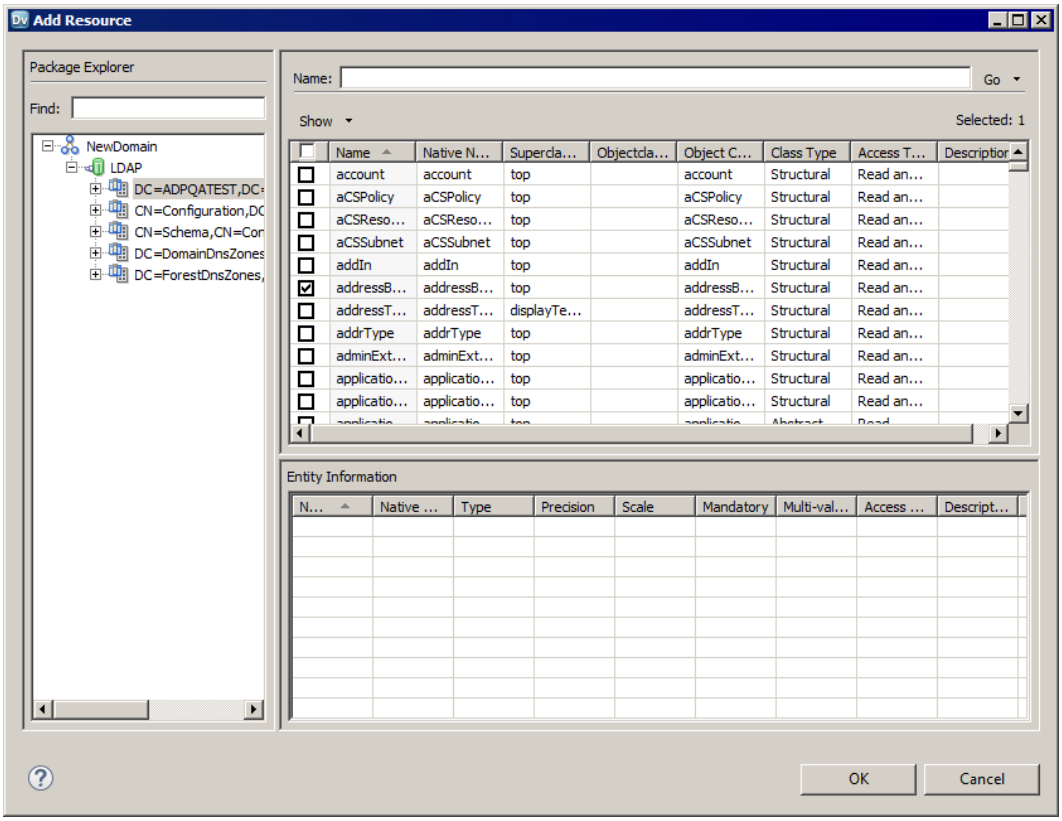
   The following image shows the **New LDAP Data Object** dialog box:



4. In the **New LDAP Data Object** dialog box, click **Browse** to choose an LDAP connection.
5. In the **Choose Connection** dialog box, choose the LDAP connection, and click **OK**.

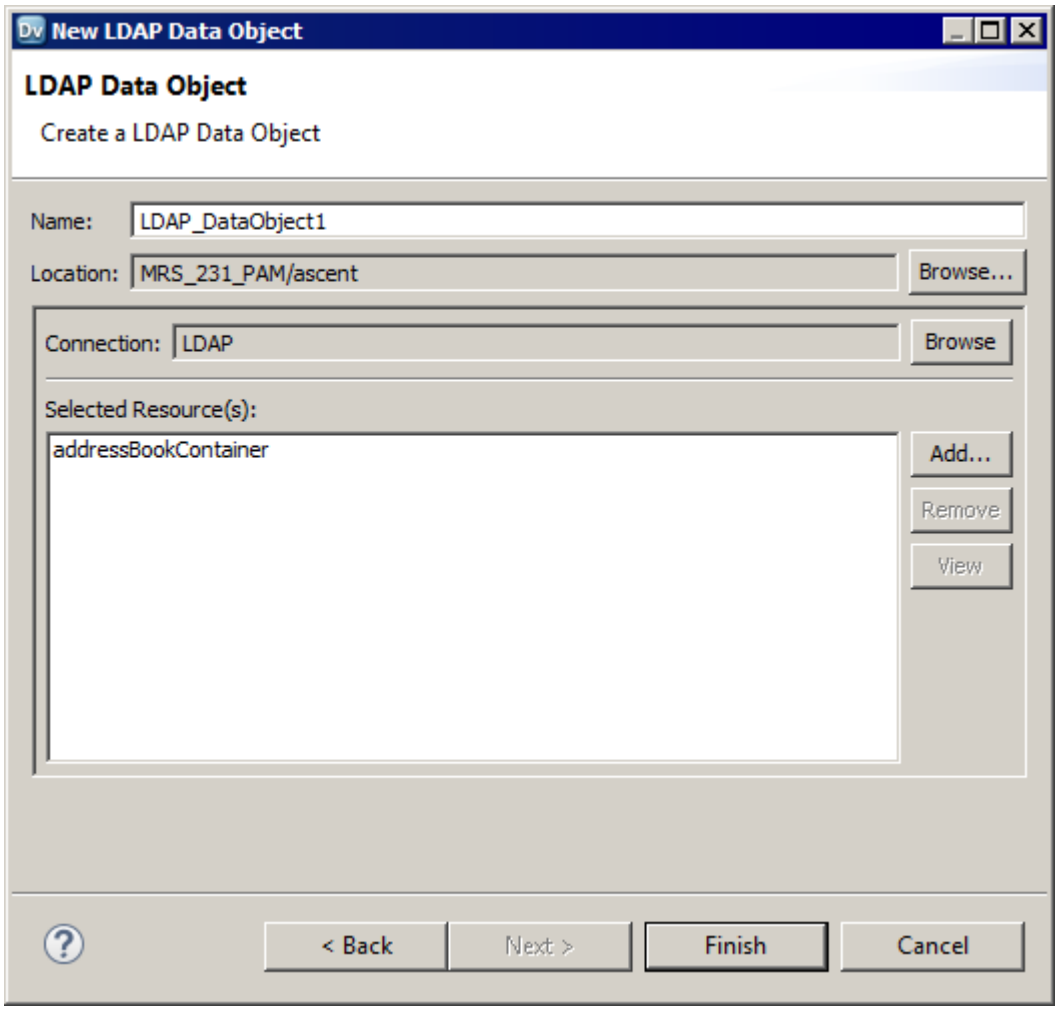
   The new LDAP connection appears in the **New LDAP Data Object** dialog box.
6. To select an LDAP data source, click **Add**.

   The following image shows the **Add Resource** dialog box:



7. In the **Add Resource** dialog box, select an LDAP data source, and click **OK**.

   The following image shows the **New LDAP Data Object** dialog box with connection and resource details:



8. Click **Finish**.

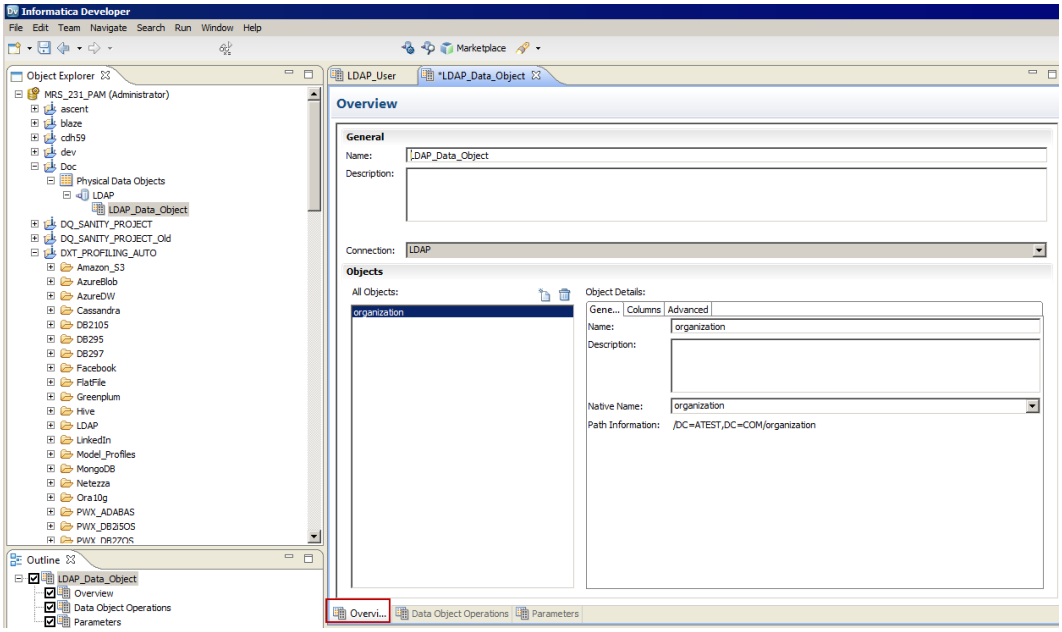
   The data object appears in the project folder.

Creating a Read Operation on the LDAP Data Object in the Developer Tool

To use the LDAP data object to create and run a column profile, create a read operation on the LDAP data object in the Developer tool.

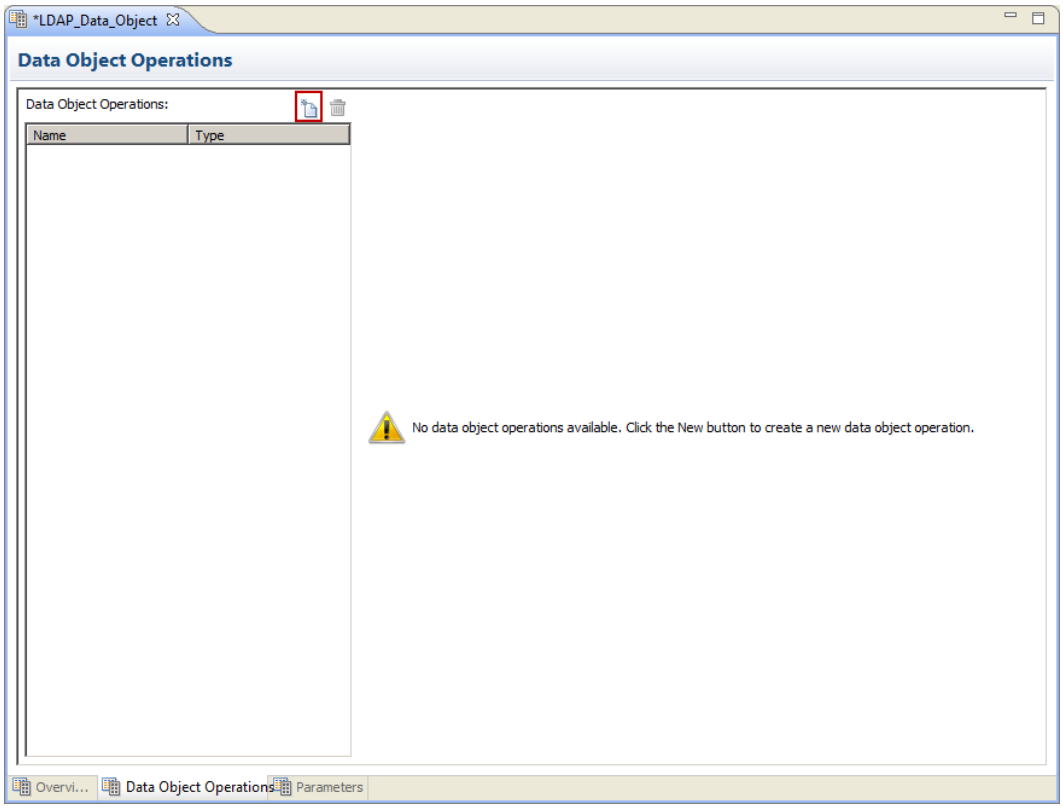
1. In the Developer tool, double-click the LDAP data object in the **Object Explorer** view.

   The following sample image shows the **Overview** tab for an LDAP data object:



2. Click the **Data Object Operations** tab.

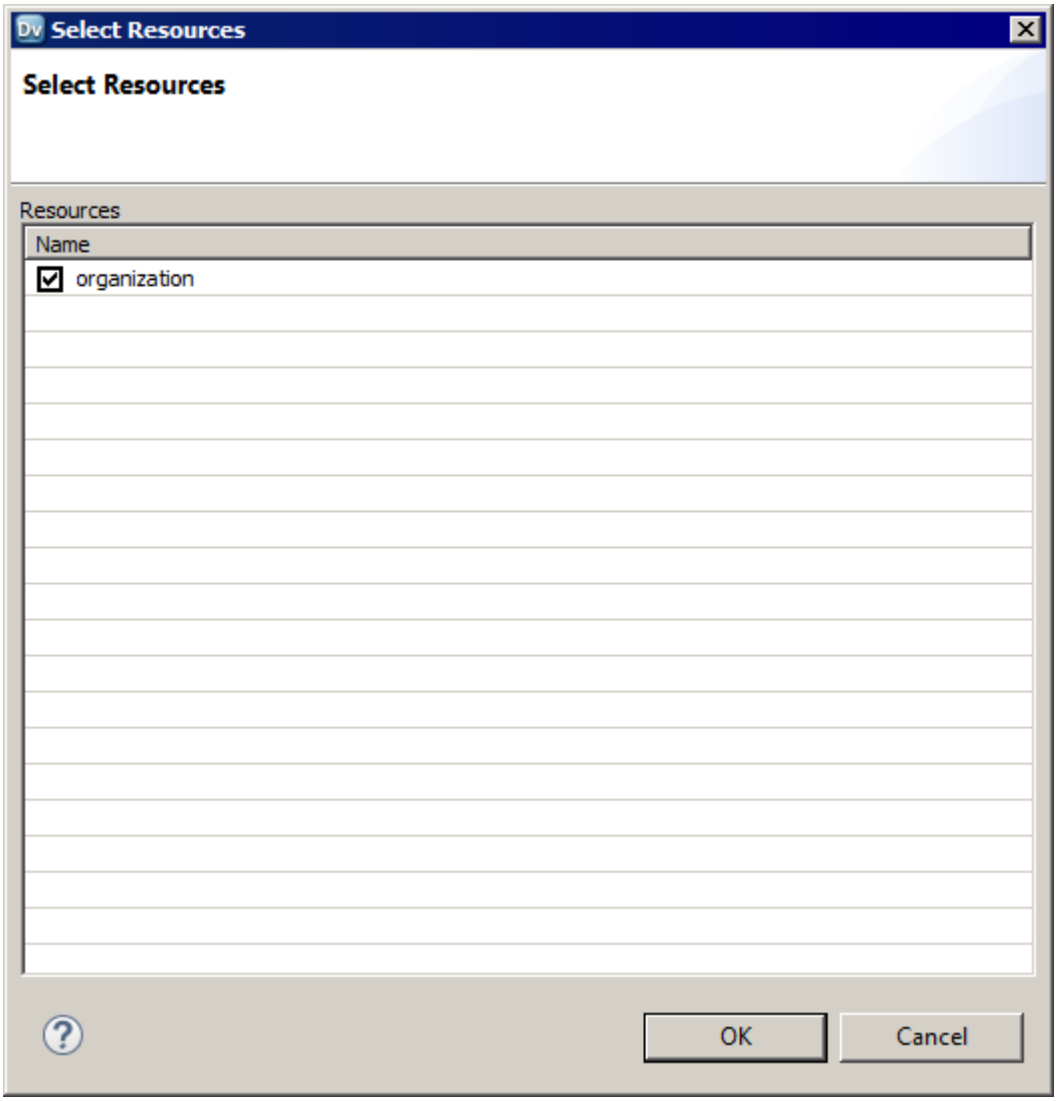
   The following sample image shows the **Data Object Operations** tab for the LDAP data object:



3. In the **Data Object Operations** tab, click the New icon ( ).

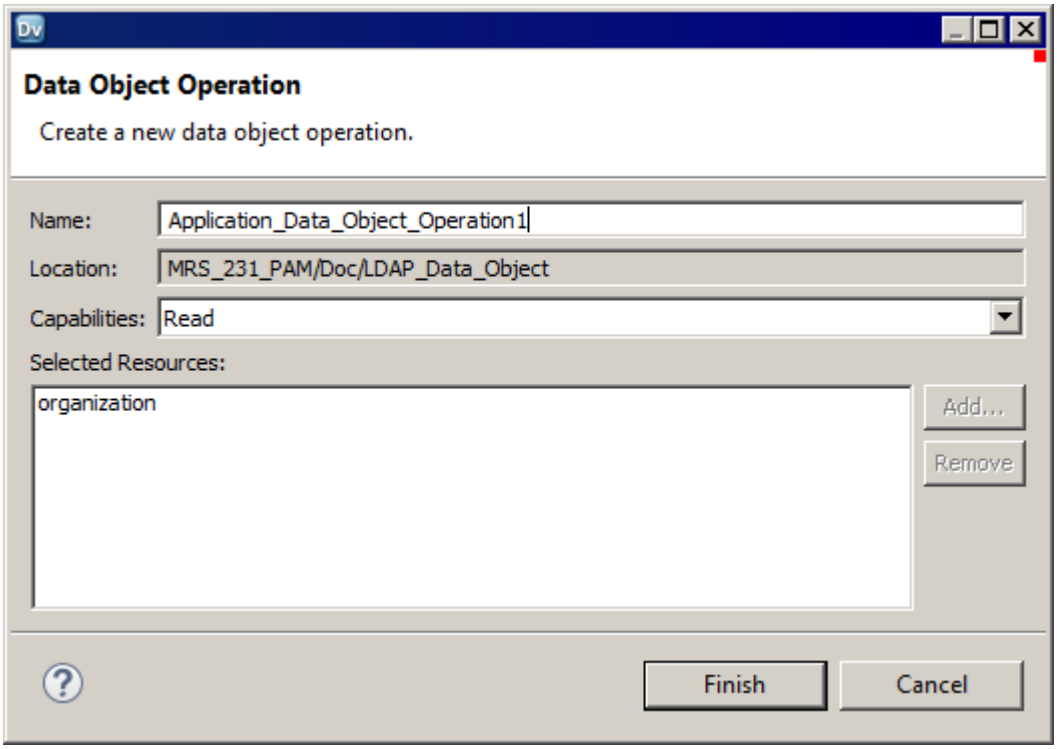
   The **Data Object Operation** dialog box appears.
4. In the **Data Object Operation** dialog box, choose the **Read** option from the list of **Capabilities**.
5. In the **Data Object Operation** dialog box, click **Add**.

   The following sample image shows the **Select Resources** dialog box:



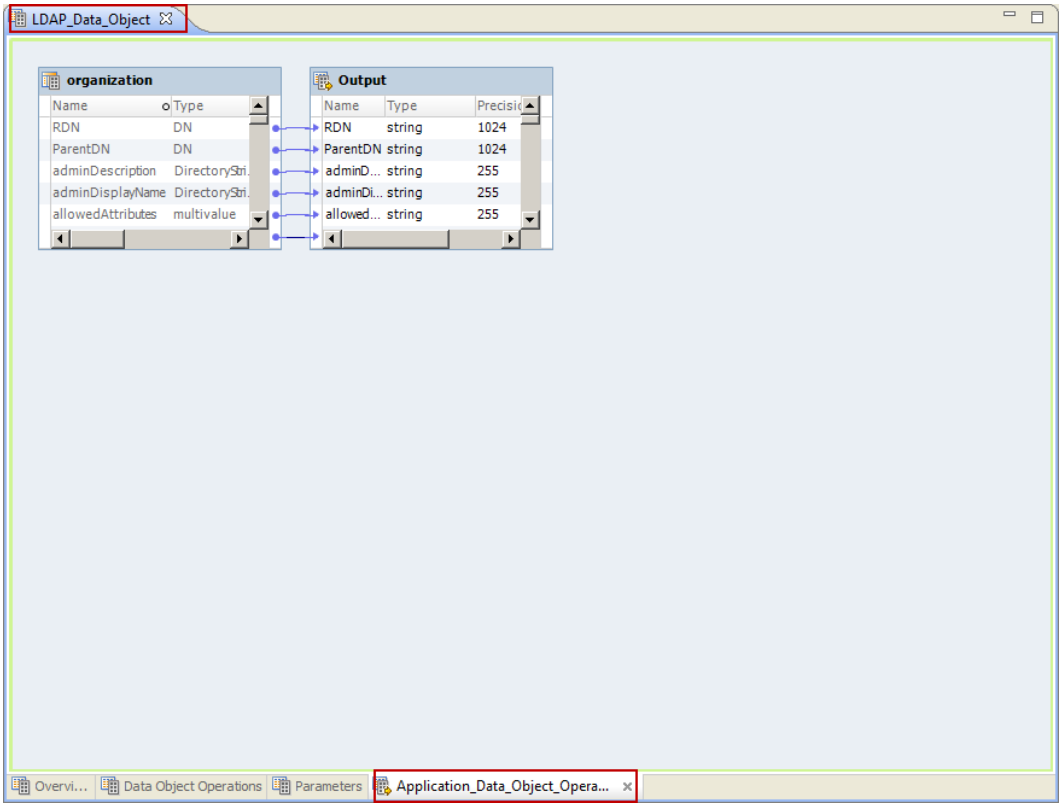
6. In the **Select Resources** dialog box, select a data object, and click **OK**.

   The following sample image shows the **Data Object Operation** dialog box:



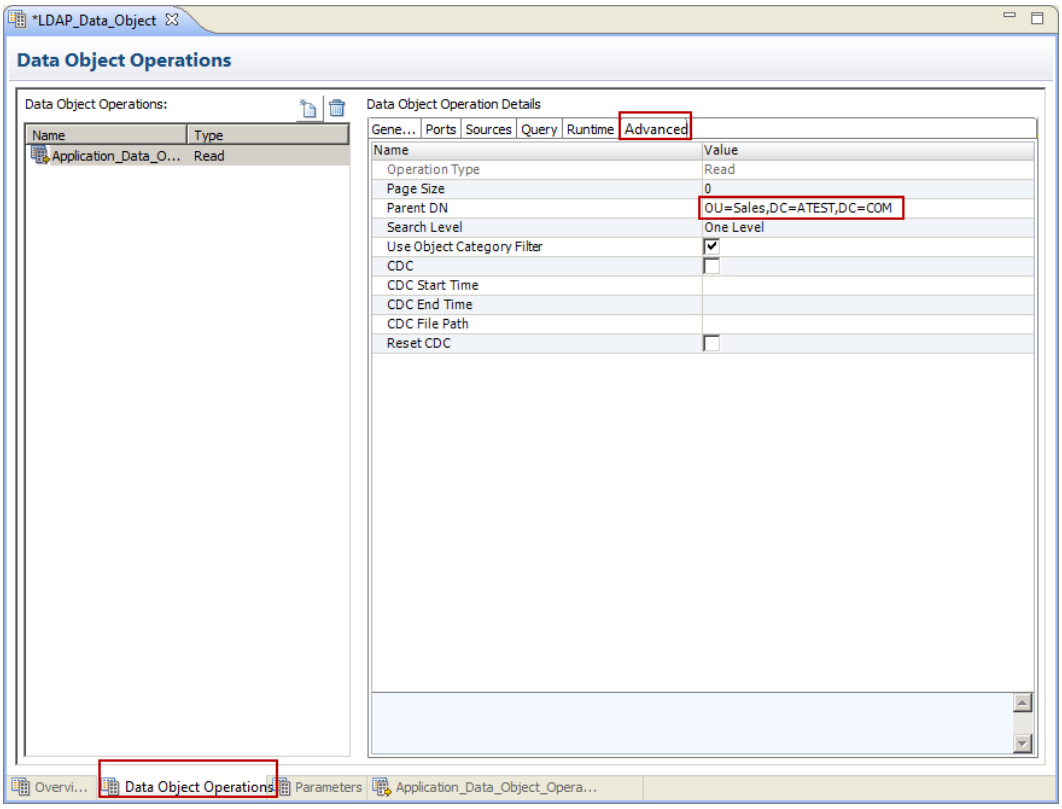
7. Click **Finish**.

   The following sample image shows the read operation for the data object in the editor:



8. In the **Data Object Operation** tab, click the **Advanced** tab.

The following sample image shows the **Advanced** tab:



9. In the **Advanced** tab, enter the **Parent DN** value, and click **File** > **Save** to save the data object properties.

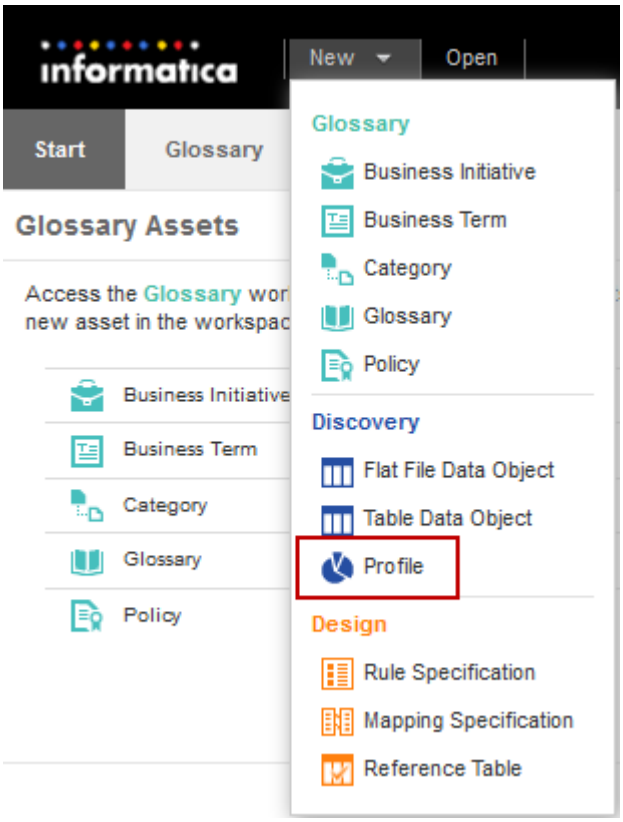
   The Parent DN is a distinguished name (DN) in an LDAP directory server namespace from where you want to fetch data.

Creating a Column Profile on the LDAP Data Object in the Analyst Tool

To analyze the data quality issues in the LDAP data source, you can create and run a column profile on the data object in the Developer tool or Analyst tool.

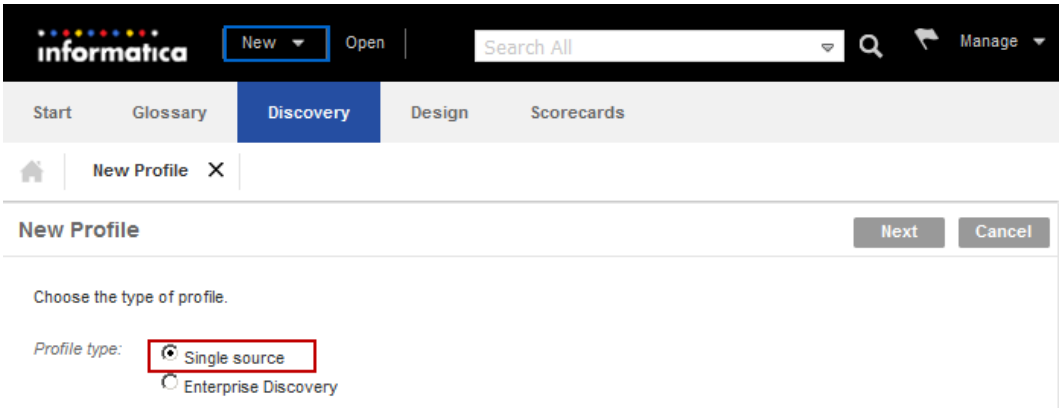
1. In the Analyst tool, click **New** > **Profile**.

   The following sample image shows the **Profile** option in the Analyst tool:



2. The **New Profile** wizard appears. Select **Single source**, and click **Next**.

   The following sample image shows the **New Profile** wizard:

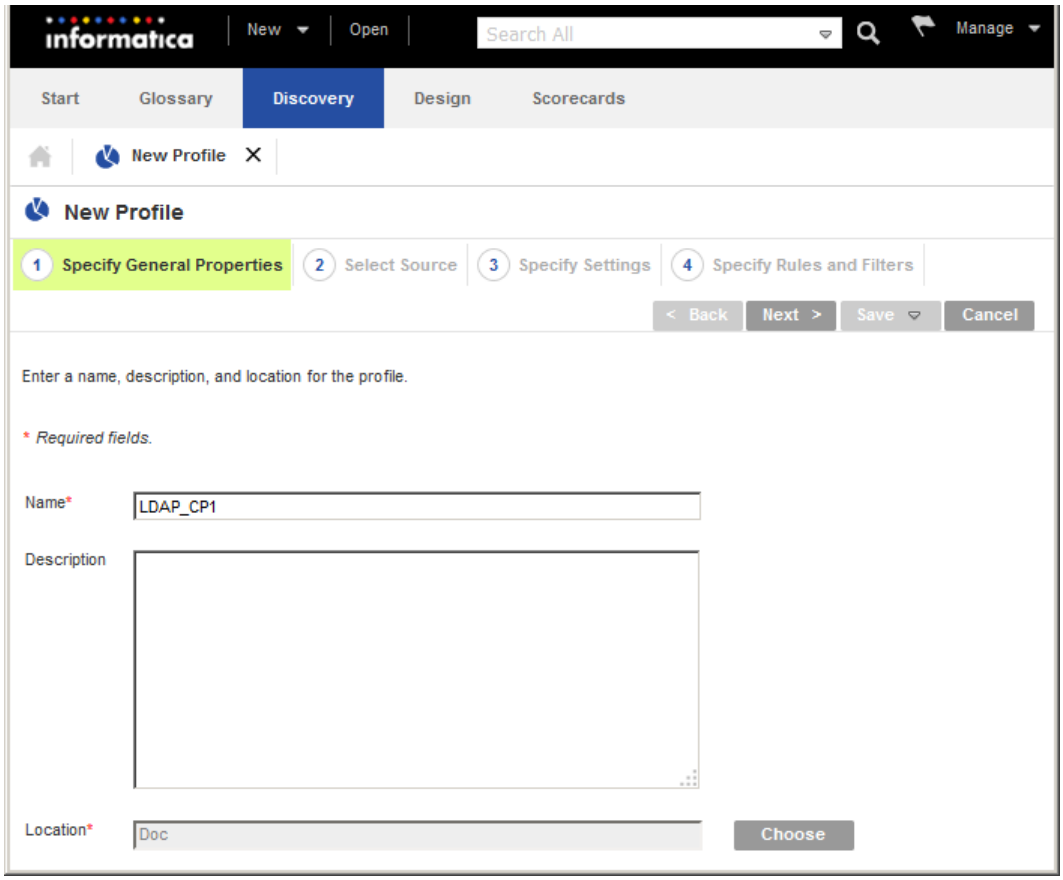


3. In the **Specify General Properties** screen, enter the following details:

   - Name. Enter a name for the profile.

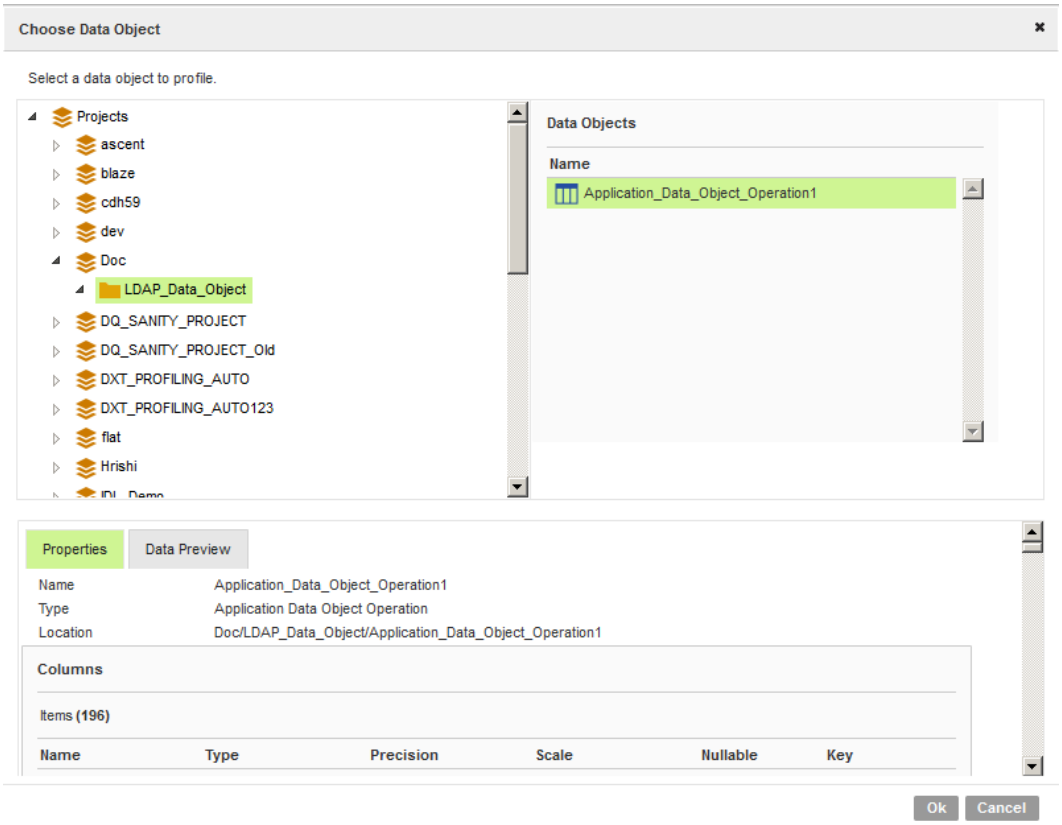
- Location. Select the project or folder where you want to create the profile.
- Description. Optional, you can enter a description for the profile.

The following sample image shows the **Specify General Properties** screen in the **New Profile** wizard:



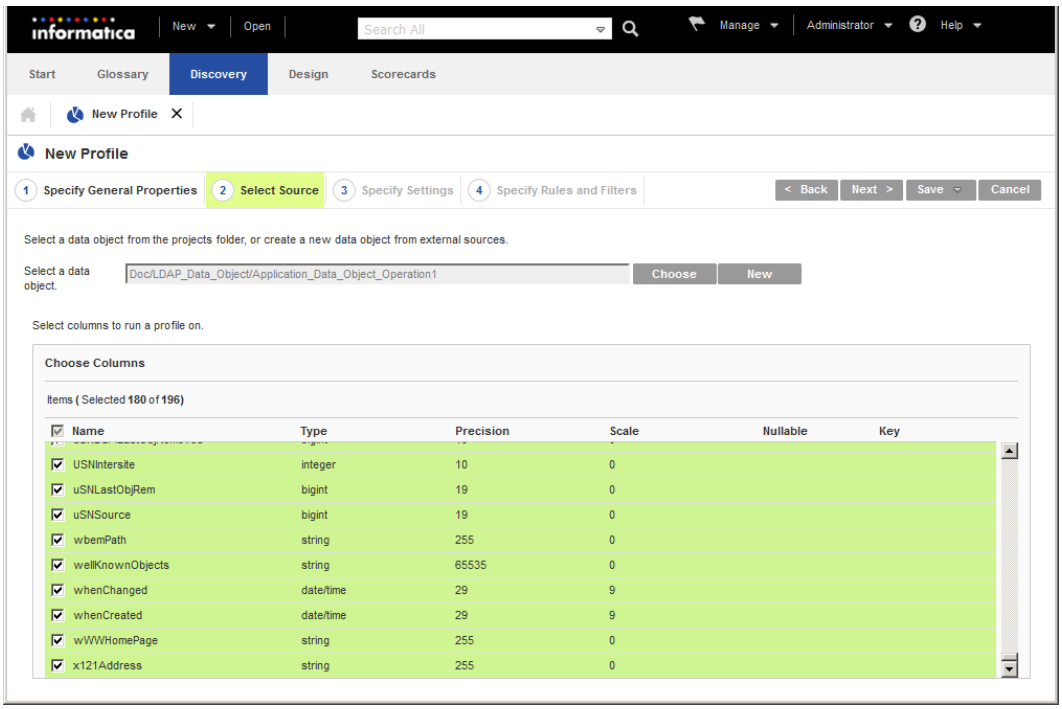
4. Click **Next**.
5. In the **Select Source** screen, click **Choose** to choose an LDAP data object.

   The following sample image shows **Choose Data Object** dialog box:



6. In the **Select Source** screen, click **Next**.

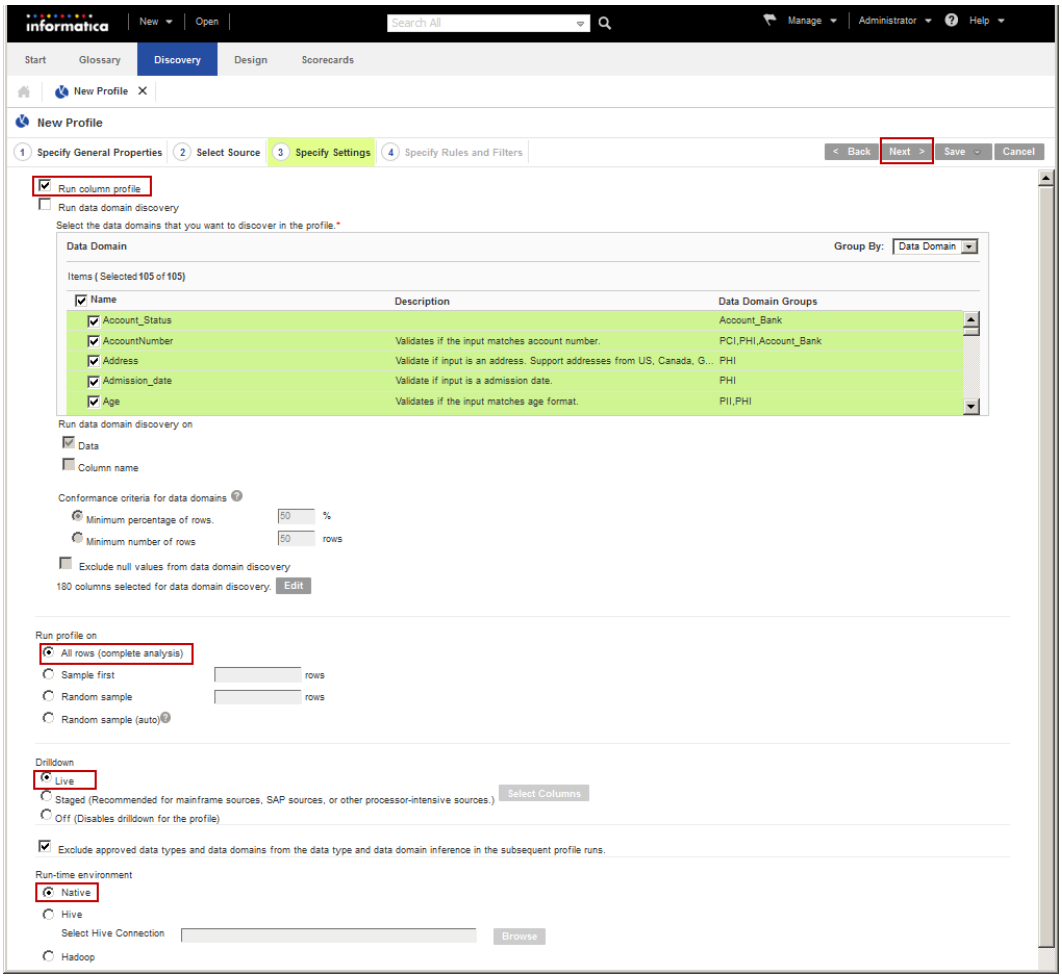
   The following sample image shows the **Select Source** screen in the **New Profile** wizard:



7. In the **Specify Settings** screen, choose the following options:

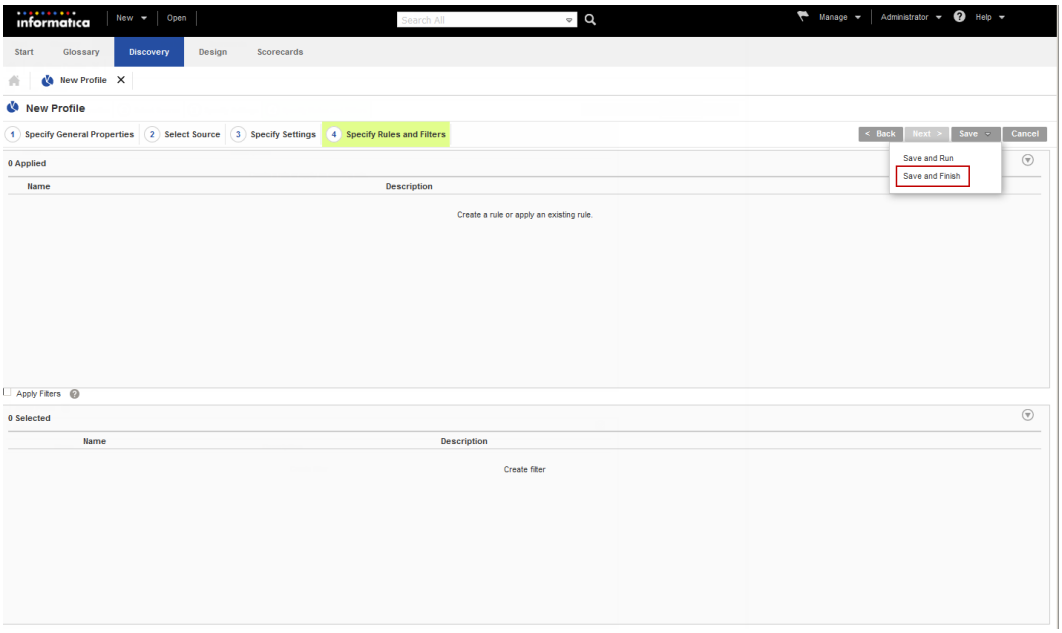
- **Run column profile** option.
- **All rows** option as the sampling option.
- **Live** option to drill down on live data to read the current data in the data source.
- **Native** option as the run-time environment.

The following sample image shows the **Specify Settings** screen in the **New Profile** wizard:



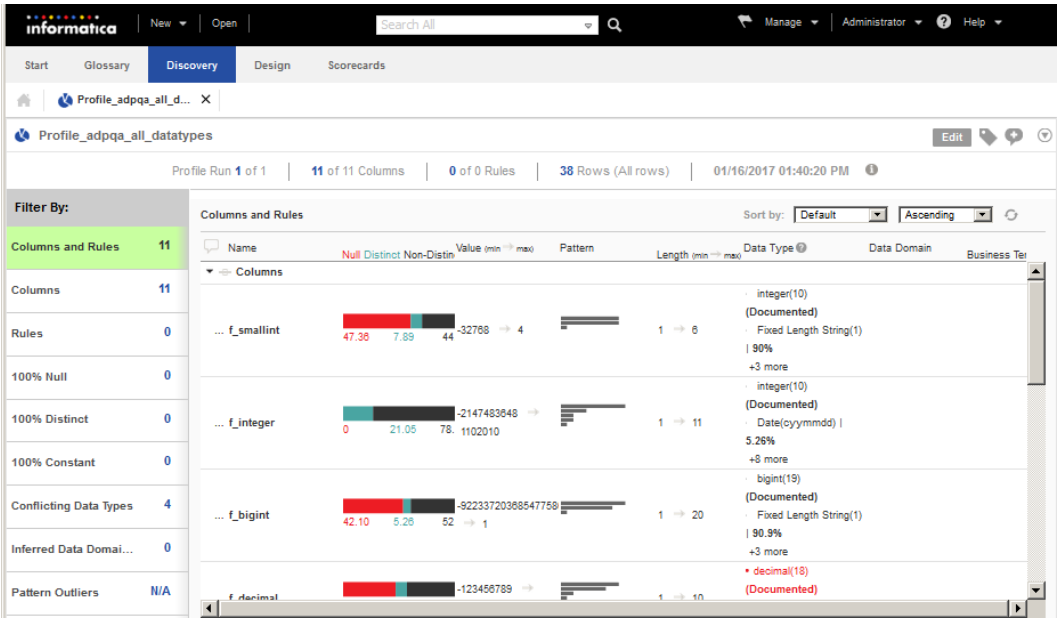
8. Click **Next**.
9. In the **Specify Rules and Filters** screen, click **Save and Run** to save and run the column profile.

   The following sample image shows the **Save and Run** option in the **Specify Rules and Filters** screen:



10. The profile results appear in summary view. Click each column to view column-specific profile results in the detailed view.

    The following sample image shows a sample summary view of the profile results:



    You can open the profile in the Developer tool and view the profile results.

    You can also create a column profile on an LDAP data object in the Developer tool and view the profile results in the Developer tool and Analyst tool.

    For more information about column profiles, see the *Data Discovery Guide*.

Author

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